Spot Light Biriyani & POS Suite - Master Guide, Diagrams & Business Playbook

1. System Architecture & Tech Stack

```
graph TD
    subgraph Client_Browser ["Client Browser / Counter Tablet"]
        UI["React 19 + TanStack Router UI"]
        subgraph Public_Site ["Public Website (Customer Facing)"]
            Home["/ (Hero, Live Handi Status, WhatsApp Modal)"]
            Menu["/menu (Categorized Dishes & Prices)"]
            Branches["/branches (Tharamani, Adyar, Velachery)"]
            Contact["/contact (Bulk & Catering)"]
        end
        subgraph POS_App ["Spot Light Manager (/app)"]
            Dash["/app (KPIs, Profit Margin, Charts)"]
            Bill["/app/billing (POS Counter, Swiggy, Zomato, Custom Dishes)"]
            Rep["/app/reports (Item breakdown, Prep Forecast)"]
            Stock["/app/stock (Inventory & Waste Movements)"]
            Items["/app/items (Menu & Plate Cost Editor)"]
            Set["/app/settings (Channels, Backup/Restore)"]
        end
        Store["POS Store Engine (src/lib/pos-store.ts)"]
        LocalStorage[("Browser LocalStorage\nKey: slb.pos.v1\n100% Offline Capable")]
    end

    subgraph Peripherals ["Hardware & Integrations"]
        Printer["80mm Thermal Receipt Printer (USB / Bluetooth)"]
        Audio["Web Audio API (Synthesizer Chimes)"]
        WhatsApp["WhatsApp Business Cloud Link"]
    end

    Home --> Store
    Menu --> Store
    Dash --> Store
    Bill --> Store
    Rep --> Store
    Stock --> Store
    Items --> Store
    Set --> Store

    Store <--> LocalStorage
    Bill -->|Browser Print Dialog| Printer
    Bill -->|Sound Chime| Audio
    Home -->|Direct Order| WhatsApp
```

2. Complete Entity Relationship Data Model (ERD)

```
erDiagram
    BRANCH ||--o{ SALE : "records"
    BRANCH ||--o{ STOCK_ITEM : "stores"
    CHANNEL ||--o{ SALE : "routes"
    ITEM ||--o{ SALE_LINE : "ordered_in"
    SALE ||--|{ SALE_LINE : "contains"
```

```
STOCK_ITEM ||--o{ STOCK_MOVE : "tracks"

BRANCH {
    string id PK "tharamani, adyar, velachery"
    string name "Tharamani"
    string address "Kazhikundram, OMR"
    string phone "+91 90000 00001"
}

CHANNEL {
    string id PK "counter, swiggy, zomato"
    string name "Swiggy"
    number commissionPct "22.0%"
    number packingCost "■8.00"
}

ITEM {
    string id PK "cb-full, mb-full, kushka"
    string name "Chicken Biriyani (Full)"
    string category "biryani, starters, breads"
    number price "Counter price ■180"
    number onlinePrice "Swiggy price ■220"
    number cost "Plate food cost ■96"
    boolean active "true/false"
}

SALE {
    string id PK "sale-uuid"
    string branchId FK
    string channel FK "counter / swiggy / zomato"
    number gross "■540"
    number discount "■20"
    number net "■520 (Customer pays)"
    number cogs "■236 (Food cost)"
    number commission "■114.40 (Swiggy cut)"
    number packing "■8.00"
    number profit "■161.60 (Net profit)"
    string payment "cash / upi / card / online"
    datetime timestamp
}

SALE_LINE {
    string itemId FK
    string name "Chicken Biriyani (Full)"
    number qty "2"
    number price "■180"
    number cost "■96"
}

STOCK_ITEM {
    string id PK "st-chicken, st-rice, st-oil"
    string name "Broiler Chicken"
    string unit "kg / liter / packet"
    number onHand "42.5"
    number reorderLevel "15.0"
    number costPerUnit "■160/kg"
}

STOCK_MOVE {
    string id PK "move-uuid"
    string stockItemId FK
    string type "purchase / used / wastage"
    number qty "25"
    string note "Morning market purchase"
```

```
        datetime timestamp
    }
```

3. Order & True Profit Calculation Flowchart

```
flowchart TD
    A["Cashier Selects Channel:\nCounter / Swiggy / Zomato"] --> B["Select Dishes (Full, Half, 65)
 & Multi-Quantities\nor Click '+ Custom / Open Dish'"]
    B --> C["Gross Revenue = Σ (Price × Qty)"]
    C --> D["Apply Discount (■)\nNet Customer Revenue = Gross - Discount"]

    subgraph Deductions ["Operating Cost Deductions"]
        E["Food Cost (COGS) = Σ (Plate Cost × Qty)"]
        F["Platform Cut = Net Revenue × Channel % (e.g. Swiggy 22%)"]
        G["Packaging Fee = Fixed Order Fee (e.g. ■8.00)"]
    end

    D --> Deductions
    Deductions --> H["NET PROFIT FORMULA:\nProfit = Net Revenue - COGS - Commission - Packing"]
    H --> I{"Live Margin Check"}
    I -->|Healthy Margin| J["■ Green Highlight on POS Screen\nReal-time Margin % Displayed"]
    I -->|Low Margin Alert| K["■ Warning Indicator"]
    J --> L["Cash Tendered Input:\n[Exact] [■200] [■500] [■1000] -> Calculates Return Change"]
    L --> M["Save Sale -> LocalStorage\nPlay Web Audio Chime (Ding!)\nPrint 80mm Receipt with Toke
n #"]
```

4. End-to-End Daily Restaurant Workflow

```
sequenceDiagram
    autonumber
    actor Staff as Restaurant Staff / Cashier
    actor Owner as Restaurant Owner
    participant POS as Spot Light Manager
    participant Kitchen as Kitchen Dum Master
    participant Printer as 80mm Thermal Printer

    Note over Staff,Kitchen: MORNING (11:00 AM - 12:00 PM)
    Staff->>POS: Open /app/stock -> Record Morning Purchases (Chicken, Rice, Oil)
    POS->>POS: Calculate Updated Inventory & Stock Value
    Staff->>POS: Open /app/reports -> Check "Prep Tomorrow" forecast plates
    Staff->>Kitchen: Handover Dum Handi cooking targets (e.g., 65 Chicken, 30 Mutton)

    Note over Staff,Kitchen: LUNCH & DINNER RUSH (12:30 PM - 10:30 PM)
    Staff->>POS: Open /app/billing -> Choose Channel (Counter / Swiggy / Zomato)
    Staff->>POS: Tap dishes -> Enter discount -> Select Cash/UPI
    POS->>Staff: Shows live true profit & cash change to return
    Staff->>POS: Tap "SAVE BILL & PRINT"
    POS->>POS: Plays cash register chime (Ding!)
    POS->>Printer: Output 80mm receipt with Token #039 & Kitchen Note
    POS->>POS: Automatically deducts ingredients from stock

    Note over Staff,Owner: NIGHT AUDIT & CLOSING (10:45 PM)
    Owner->>POS: Open /app Dashboard -> Check Today's Net Revenue, Net Profit, Swiggy Cut
    Owner->>POS: Open /app/reports -> Export Day CSV for Accountant
    Owner->>POS: Open /app/settings -> Download JSON Backup (Weekly to Google Drive)
```

5. Local Network & Mobile Phone Architecture

```
graph TD
    Laptop["Your Laptop (Windows PC)<br/>Running Server on Port 8080<br/>Local IP: 10.223.36.25"]
    Router["Local Wi-Fi Router or Mobile Hotspot<br/>(Same Local Network Subnet)"]
    Phone["Your Mobile Phone or Counter Tablet<br/>(Chrome or Safari Browser)"]

    Laptop <-->|Connected to Local Wi-Fi| Router
    Router <-->|Connected to Same Wi-Fi| Phone

    Phone -->|Step 1: Type http://10.223.36.25:8080| Laptop
    Laptop -->|Step 2: Streams Live POS App| Phone
```

6. Directory Layout of the Master Codebase

```
spotlight-biriyani/
■■■ public/                      Static web assets
■   ■■■ favicon.svg              Glowing brand flame SVG icon
■   ■■■ favicon.ico              32x32 browser tab icon
■
■■■ src/
■   ■■■ routes/                  TanStack Router (File-based routes)
■   ■   ■■■ __root.tsx           Master Shell (Fonts, Favicons, Sonner toaster)
■   ■   ■■■ index.tsx            Homepage (Live Handi banner, WhatsApp modal, Buckets, Reviews)
■   ■   ■■■ menu.tsx             Public menu (Reads live prices from POS store)
■   ■   ■■■ branches.tsx         Kitchen locations with Google Maps directions
■   ■   ■■■ contact.tsx          Bulk catering inquiry form
■   ■   ■
■   ■   ■■■ app.tsx              POS App Shell (Branch dropdown, navigation tabs)
■   ■   ■■■ app.index.tsx        Manager Dashboard (Revenue, Food Cost, Net Profit %)
■   ■   ■■■ app.billing.tsx      Fast touch billing terminal, multi-item counters, custom items, r
eceipts
■   ■   ■■■ app.reports.tsx      Item sales counts, P&L breakdown & Prep Forecast
■   ■   ■■■ app.stock.tsx        Raw materials (Chicken, Rice, Oil) on hand & wastage
■   ■   ■■■ app.items.tsx        Menu & Cost Editor (Counter price, Swiggy price, Plate cost)
■   ■   ■■■ app.settings.tsx     Branch names, Swiggy/Zomato commission %, JSON Backup
■   ■
■   ■■■ lib/
■   ■   ■■■ pos-store.ts         THE CORE ENGINE: Database, profit math, storage, demo seed data
■   ■
■   ■■■ components/site/
■   ■   ■■■ SiteShell.tsx        Public website navigation header & footer
■   ■
■   ■■■ assets/                  Food photography (Biriyani, Mutton, Kushka)
■   ■■■ styles.css               All design tokens (Embers & Charcoal OKLCH color palette)
■
■■■ docs/
    ■■■ user-manuals/            Dedicated Step-by-Step Operating Manuals
    ■   ■■■ 1_RESTAURANT_OWNER_MANUAL.md
    ■   ■■■ 2_MANAGER_AND_CASHIER_MANUAL.md
    ■   ■■■ 3_DEVELOPER_AND_ADMIN_A_TO_Z_GUIDE.md
    ■   ■■■ 4_WHITE_LABEL_REBRANDING_GUIDE.md
    ■   ■■■ printable-pdf-html/ (Printable HTML files ready to save as PDF via Ctrl + P)
    ■
    ■■■ delivery/                14 Client proposals, pricing sheets, sales pitch decks (English,
Tamil, Hindi)
```

7. Dedicated User Manuals & Printable PDF Files

All manuals are organized inside [docs/user-manuals/](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals):

1. **[1_RESTAURANT_OWNER_MANUAL.md](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals/1_RESTAURANT_OWNER_MANUAL.md)**

- *For the Owner*: Daily morning to night routine, how to check true profit, stop Swiggy leakage, and download Sunday backups.

- *Printable PDF*: [1_RESTAURANT_OWNER_MANUAL.html](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals/printable-pdf-html/1_RESTAURANT_OWNER_MANUAL.html)

2. **[2_MANAGER_AND_CASHIER_MANUAL.md](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals/2_MANAGER_AND_CASHIER_MANUAL.md)**

- *For the Cashier / Counter*: Step-by-step billing, ordering multiple dishes (1 full, 1 half, 1 65), adding custom off-menu dishes, using quick cash tender chips, and printing 80mm thermal receipts with token numbers.

- *Printable PDF*: [2_MANAGER_AND_CASHIER_MANUAL.html](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals/printable-pdf-html/2_MANAGER_AND_CASHIER_MANUAL.html)

3. **[3_DEVELOPER_AND_ADMIN_A_TO_Z_GUIDE.md](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals/3_DEVELOPER_AND_ADMIN_A_TO_Z_GUIDE.md)**

- *For You (Developer / Admin)*: Starting/stopping server, finding your Wi-Fi IP, opening on mobile phones, free cloud deployment on Vercel/Netlify.

- *Printable PDF*: [3_DEVELOPER_AND_ADMIN_A_TO_Z_GUIDE.html](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals/printable-pdf-html/3_DEVELOPER_AND_ADMIN_A_TO_Z_GUIDE.html)

4. **[4_WHITE_LABEL_REBRANDING_GUIDE.md](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals/4_WHITE_LABEL_REBRANDING_GUIDE.md)**

- *For Business Scaling*: How to adapt this software for ANY restaurant (Chai cafe, burger joint, Mandi shop) in 15 minutes, pricing models, pitch scripts, and financial models.

- *Printable PDF*: [4_WHITE_LABEL_REBRANDING_GUIDE.html](file:///c:/Users/ManikandanSoundar/OneDrive%20-%20Adeahub/Personal%20Work/spotlight-biriyani-full-kit/spotlight-biriyani/docs/user-manuals/printable-pdf-html/4_WHITE_LABEL_REBRANDING_GUIDE.html)